

PROFESSIONAL SUMMARY

Principal Engineer with nearly 10 years of experience architecting high-impact mobile solutions used by millions across FinTech, banking, travel, and Web3. Expert in Swift, SwiftUI, Clean Architecture, payment systems, analytics, the SDLC, and performance optimization. Led engineering teams, mentored 10+ engineers, and established technical best practices. Skilled in AI and LLM integration, including Claude, ChatGPT, and MCP, with a track record of shipping 14+ production apps and managing projects worth over \$2 million.

CORE COMPETENCIES

- iOS Architecture and Design: SOLID Principles, Modular Architecture, Design Systems, Mobile Architecture, MVC, MVVM & VIPER, Apple Connectivity, Apple Developer Console, Reusable Frameworks with Xcode.
- Mobile Platform Engineering: Swift 3/4/5/6, Objective-C, Production SwiftUI, UIKit, Cocoa Touch, Auto Layout, Swift Package Manager, CocoaPods, State Driven Navigation, Deep Linking, Universal Links, AVFoundation, Core Graphics, Interface Builder.
- Concurrency and Performance: Concurrency (Async/Await), Combine, GCD, Threading Models, Memory Optimization, Instruments Profiling, Performance Tuning, App Startup Time Optimization, Application Stability, Frozen Frames Prevention.
- Networking and Integration: URLSession, REST APIs, Offline-first, Web Services Integration, SOAP, WebSocket, GraphQL, Socket Programming, API Design, Caching, Retry Mechanisms, Secure Data Exchange.
- FinTech and Payments: Apple Pay, Stripe, Plaid, Coinbase, WalletConnect, UPI, PayPal, Credit Card Processing, In-App Purchase, 3D Secure and Digital Assets.
- Analytics and Observability: Analytics (Firebase, Adobe), Insider SDK, Crash Analysis & Monitoring, Performance Monitoring, Event Tracking, Operational Insights.
- Testing and Quality Engineering: Unit Testing, Snapshot Testing, TDD, XCTest, Test Coverage, Debugging, Code Quality, Swift Lint, Technical Debt Reduction, Release Readiness and Wireshark.
- Cloud, CI/CD and DevOps: CI/CD Pipelines, Git, Build Automation, Release Management, Feature Flags, Remote Configuration, Continuous Delivery, Provisional Profiles, App Store Deployment, App Store Connect.
- Product Domains: FinTech, Banking, Digital Payments, Web3/Crypto, Travel and Booking Platforms, Donation Platforms, Enterprise Applications, TestFlight.
- Mobile Engineering Leadership: Technical Leadership, Problem Solving, Decision Making, Architecture Reviews, Code Reviews, Mentorship, Agile/Scrum, Stakeholder Management, Cross Functional Collaboration, Product & Feature Delivery, Management (Team, Project, Risk, Budget & Risk), SDK architecture, Conflict Resolution, Product Strategy, Cost Optimization.
- AI and Developer Productivity: OpenAI Codex, ChatGPT, Claude, Cursor AI, GitHub Copilot, Gemini, AI/ML assisted Development, Code Generation, Refactoring, Unit Test Generation, Documentation, Debugging.
- BLE communication: BLE Layer, device pairing, authentication, synchronization, OTA updates, and real time data streaming, device controls, and sensor data access, design wellness features such as heart rate, SpO₂, sleep, activity, GATT, Wearable, HealthKit, nRF Connect and stress monitoring.
- Security and Authentication: Data encryption and mobile security, including AES-256, RSA, ECC, digital signatures, tokenization, TLS/HTTPS, SSL pinning, mTLS, Perfect Forward Secrecy (PFS), data at rest/in transit protection, JWT, OAuth 2.0, bearer/access/refresh tokens, Face ID, Touch ID, Web Auth, MITM attack prevention, public/private and secret key management, jailbreak detection, FIDO, PCI DSS, SOC 2, Apple Keychain, Secure Enclave, Passkeys, secure storage, authentication flows, OTP verification, and payment security.

PROFESSIONAL EXPERIENCE

Principal Engineer | BeneFi Global | Delhi, In (Remote For USA)
| (Promoted)

Aug 2024 – Present
July 2026

Staff Engineer and Mobile Lead

Official NP: 14 days

- Architected extensible fiat, stock, and crypto payment solutions, compliance, scalability, and future platform extensibility.
- Built blockchain enabled self-custody wallets and digital asset transaction infrastructure.
- Integrated Stripe, Plaid, Coinbase, and WalletConnect for secure FinTech capabilities.
- Introduced MVVM, and Protocol Oriented Programming to establish engineering standards.
- Architected end to end user journeys including authentication, onboarding, identity verification (IDV) and state driven navigation.
- Design secure networking infrastructure with URLSession, API abstractions, caching, and retry strategies.
- Implemented token lifecycle management, feature flags, remote configuration, and observability across the application.
- Significantly improved maintainability, testability, development velocity, and long-term platform scalability.
- Implemented analytics, observability, and crash monitoring to improve platform reliability and engineering excellence.
- Applied performance optimization and reusable engineering practices to ensure stability.
- Delivered data driven mobile experiences that accelerated feature delivery, improved application quality, and enhanced user engagement.
- Optimized application performance using Instruments (Memory, CPU, and I/O) and mentored engineers on best practices.

- Led architecture reviews, technical design discussions, and code reviews across the engineering organization.
- Utilized Claude Code for intelligent code analysis, security pattern detection, and architecture optimization across payment and blockchain modules.
- Collaborated with Product, Design, Backend, QA, and leadership teams to drive technical strategy and successful product delivery.

SDE and iOS Lead | Cleartrip Pvt. Ltd. | (Gurugram Hybrid)

Feb 2024 – Jun 2024

- Led iOS architecture, system design, and technical roadmap for Cleartrip Middle East and Flyin applications.
- Drove scalable, modular, and production ready solutions while partnering with engineering leadership on technical strategy and release planning.
- Reduced application crashes by 15% through systematic analysis and memory Management.
- Optimized API integrations and refactored legacy modules, significantly reducing technical debt and strengthening release confidence.
- Leveraged AI for debugging, performance optimization, rapid prototyping, technical documentation, and architecture decision Records.
- Architected and delivered mission critical travel capabilities including booking flows, and GraphQL integration.
- Built secure payment workflows with Apple Pay, Credit Cards, 3D Secure, encryption, and secure storage.
- Implemented analytics, operational reporting, and baggage & passenger insurance features, improving user engagement, conversion, and transaction reliability.
- Owned technical delivery from architecture through production, leading architecture reviews, code reviews, and mentoring engineers.
- Collaborated cross functionally with Product, Backend, QA, Design, and business stakeholders to ensure production readiness and successful market launches.
- Design reusable platform components and internal frameworks, improving development velocity, application consistency, and release quality.
- Enabled faster feature delivery across both high traffic travel applications through reusable components and standardized practices.

iOS Consultant | Emirates National Bank of Dubai (Via Spare IT)

Aug 2023 – Feb 2024

- Developed scalable iOS banking modules for the ENBDX platform, Cocoa Touch, and reusable private frameworks, improving code quality and maintainability
- Integrated Insider SDK for rich push notifications, delivering personalized transaction updates, feature announcements, and customer engagement campaigns.
- Implemented Adobe Analytics across the core platform to track user journeys, funnel drop offs, feature adoption, conversion metrics, and product usage insights.
- Designed reusable private libraries for networking and shared business logic while implementing multilingual support with English and Arabic (RTL) localization.
- Enhanced the banking experience through Core Animation, interactive UI, performance optimization, and continuous feature enhancements for a stable production application.

Senior iOS Developer | Paytm Payments Bank Limited

Jan 2022 – Aug 2023

- Designed and delivered the Add Money module, improving payment workflows and transaction reliability.
- Engineered secure Wallet features, enhancing digital payment and account management experiences.
- Architected and implemented **Rupay Credit Card UPI** onboarding and payment integration with the UPI ecosystem.
- Developed Aadhaar OTP based UPI MPIN setup, ensuring secure authentication and regulatory compliance.
- Applied Unit Testing and clean code practices to improve application quality and maintainability.
- Leveraged the Backbase Mobile SDK to build scalable Banking applications, enterprise grade digital banking solutions.

Senior iOS Developer | Birlasoft Limited

Feb 2021 – Jan 2022

- Developed and enhanced iOS modules, Cocoa Touch, and Objective C, including refactoring private Cocoa Pods to improve maintainability and code quality.
- Designed and implemented an offline peer to peer synchronization system where crew devices connected to an onboard Wi Fi router to share itinerary, order, and payment updates in real time without internet connectivity.
- Built key airline features including meal collection, voucher scanning, order correction workflows, and MileagePlus payments with seamless online and offline support for flight attendants.
- Optimized application performance through architecture improvements and code refactoring, reducing technical debt while improving stability and responsiveness.
- Collaborated closely with Product, Backend, and QA teams to deliver reliable end to end features and ensure production readiness.
- Improved cabin crew efficiency by ~20% through automated in-flight ordering, payment processing, and synchronized offline data sharing across connected devices.

iOS Developer | Aeonic Edge Pvt Ltd

Jun 2019 – Jan 2021

- Designed and developed iOS applications from scratch (Dmystifi and School ERP), Cocoa Touch, Auto Layout, SQLite, and Objective C maintenance.
- Built the Dmystifi online learning platform with live and recorded classes, video playback, learning progress tracking, and SQLite based offline content storage.
- Implemented student learning validation to ensure lessons were completed sequentially, preventing users from skipping ahead while tracking total watch time and course completion.

- Developed a real time school bus tracking system using Core Location, Apple Maps, and custom route animations, enabling parents to monitor bus locations, receive ETA updates, and get push notifications when the bus approached their pickup stop or reached its destination, improving child safety.
- Integrated multiple payment gateways, Apple QR code, interactive charts, and custom animations to enhance user experience and application functionality.
- Optimized local data persistence using SQLite for offline access, synchronization, and improved application responsiveness.

iOS Developer | Prime lite technology pvt. Ltd

May 2017 – May 2019

- Developed and published multiple iOS applications (Openkart, Aiwa Top up, LimeT, Mobilio) for the UAE and Oman markets, delivering large scale e commerce, recharge, and fintech solutions like Amazon and Flipkart.
- Designed and developed a next generation billing system for Gulf City, integrating (Core Bluetooth) BLE based POS printing to generate invoices directly from iOS devices, improving checkout speed and operational efficiency.
- Implemented multilingual support (Arabic/English), real time socket communication, and rich push notifications with image, video, and audio attachments to enhance user engagement.
- Built secure payment, mobile recharge, and money transfer features, including Apple QR code integration for seamless and secure transactions.
- Developed VoIP calling applications with custom communication libraries, dialer interfaces, OTP based authentication, contact management, and comprehensive feature testing for reliable voice communication.

EXTERNAL PROJECT

TRADEDOG is an institutional grade financial services firm structuring products using and putting resources into digital assets and aiding the new startups, like us to grow exponentially. We are pleased with the straightforwardness and the reporting model of TradeDog and the foundation that upholds the capacity for any asset, in any asset class, to turn into a digital product.

- Design UI using storyboard and auto layout.
- Rest API calling for research post on Digital currency.
- Using Socket programming for live price and chat.
- Graphical representation on diagram with live market.

EDUCATION

Master of Business Administration in IT (MBA IT), Department of Business Administration, Sikkim Manipal University | 2020

Master of Computer Applications (MCA), Department of Computer Science, GGSIPU University | 2017

Bachelor of Computer Applications (BCA), Department of Computer Science, Sikkim Manipal University | 2014

LEADERSHIP SKILL

Strategic Planning, Cross-Functional Leadership, Secure Coding Practices, Test Automation, Revenue Growth, Strategic Thinking, Prompt Engineering, Quality Assurance, Performance Testing, Regression Testing, Data Privacy, Event-Driven Architecture, App Launch Time, App Size Optimization, Battery Optimization, Network Performance, Network conditioner, Pre-caching, Lazy Loading, Background Tasks, Large Scale System Design, On-Call Rotation, Advanced Debugging tool, Latency Optimization, Pub-Sub/Streaming Protocols, Protocol Buffers, Modern Build Systems, Top-Chart App Store Pedigree, Designing GenAI Capabilities, Agentic Workflows, Cross-Platform Awareness, Accessibility.

External support: Android, Kotlin, Java, Node JS, QA, Business Team, Azure, Oracle Cloud, AWS Amplify, Backend less, Docker, Kubernetes, Fastlane, Launch Darkly, Mix panel, Amplitude, Sentry, New Relic, Insta bug, .NET MAUI, Razor pay, Digital health, connected healthcare, remote patient monitoring, medical devices, application lifecycle management, React Native, Safe, Less, device trust, SCA, experimentation frameworks, regulator/audit exposure, Computer Vision/CV, PII, integration testing, Ruby, Fastlane, open-source contribution, React Native, Flutter, Cordova, A/B testing, Kotlin Multiplatform/KMP, XCFramework, semantic versioning, Circle CLI, Spring Boot, HTML5, XML, SQL, Docker, Kubernetes, Kafka, Microsoft Office.